

Sean Leader

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EDUCATION

California Polytechnic State University, San Luis Obispo

Sep '19 - Jun '24

- B.S. Statistics, Magna Cum Laude
 - Data Science Minor
 - GPA: 3.805
- M.S. Statistics
 - GPA: 3.811

Sep '23 - Jun '24

University of California, Irvine

Expected '29

- Ph.D. Statistics
 - GPA: 3.840

RESEARCH & WORK EXPERIENCE

Graduate Researcher, Gillen Lab at UC Irvine

Jun '24 - present

- Developing new efficient statistical methods with applications to biomarker discovery for Alzheimer's Disease
- Collaborating with UCI's Alzheimer's Disease Research Center to study data collection challenges in subpopulations with difficulties completing assessments

Forensic Statistician, Leader Law Firm

Jun '24 - present

- Investigating statistical reports in civil legal cases for statistical fraudulence

Graduate Mentor, PALiSaDS

Sep '23 - present

- Encouraging undergraduates to pursue graduate school through the Pacific Alliance for Low-income Inclusion in Statistics and Data Science

Biostatistics Researcher, UC Irvine Department of Statistics

Jul '23 - Aug '23

- Investigated attitudes about research, comparing the populations of decision-makers for those with Alzheimer's Disease & those with Down Syndrome

Textbook Contributor, Pearson

Jun '23 - Aug '23

- Designed practice problems and solutions for textbook *Data Science For All*

Data Scientist, Global Emancipation Network

Jan '23 - Jun '23

- Developed graphical networks to identify illicit activity associated with human trafficking

Data Scientist, HomeOptions

Jan '22 - Jan '23

- Integrated real estate data sourcing, aggregation, manipulation, and analysis through machine learning prediction models

Statistics Research Scholar, Frost Undergraduate Research Program

Jan '21 - Aug '21

- Converted existing codebase into reproducible format for future research
- Performed statistical analysis on student learning outcomes to understand how curriculum choice in introductory statistics relates to content retention

Data Entry Specialist, Carpet One

Jan '18 - Jun '19

- Managed work-order database, designed schema, and digitized existing data

TEACHING EXPERIENCE

Graduate Teaching Assistant, UC Irvine Department of Statistics

Sep '24 - Dec '25

- Taught several 50-student discussion sections, hosted regularly-attended office hours, and assisted during regular lecture in 450-student in *Basic Statistics* courses
- Developed supplementary course material, created exam questions, and facilitated midterm and final examinations

Statistics Tutor, Self-Employed & Cal Poly Writing & Learning Center

Aug '19 - present

- Providing tailored support for introductory through graduate level statistics
- Helped 100+ clients, ranging from high school students to PhD students

Workshop Instructor, California Polytechnic State University

Jan '23 - Jun '23

- Taught supplemental workshops for two upper-division statistics courses: *Mathematical Statistics* and *Estimation and Sampling Theory*

Instructional Student Assistant, California Polytechnic State University

Sep '20 - Dec '21

- Answered students' questions and evaluated quizzes and examinations
 - Worked with 300+ students across intro and advanced business statistics courses
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AWARDS

Honorable Mention: NSF GRFP, *National Science Foundation* (2025)

Outstanding Scholarship Award, *Cal Poly Statistics Department* (2023)

Community Leadership Award, *Cal Poly Statistics Department* (2023)

Community Leadership Award, *Cal Poly College of Science and Mathematics* (2023)

Dean's List, *Cal Poly* (F19, W20, S20, F20, W21, F21, W22, F22, W23, S23)

President's Honors List, *Cal Poly* (19-20, 22-23)

Frost Research Fellowship, *Cal Poly* (W21-F21)

WORKSHOPS

Reproducible documents, presentations, and websites with Quarto at U.S. Conference of Teaching Statistics (2023)

PUBLICATIONS

Lu, T. V., Campos, P., **Leader, S.**, Lee, X., Xu, H., Doran, E., Grill, J. D., & Lott, I. T. (2024). Comparing research attitudes in Down syndrome and non-Down syndrome research decision-makers. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*, 10(3). <https://doi.org/10.1002/trc2.12478>

Chance, B., Tintle, N., Reynolds, S., Patel, A., Chan, K., & **Leader, S.** (2022). Student performance in curricula centered on simulation-based inference. *Statistics Education Research Journal*, 21(3). <https://doi.org/10.52041/serj.v21i3.6>

PRESENTATIONS

ISI-BUDS 2023 Research Symposium: **Leader, S.**, Campos, P., Lee, X., Xu, H. “Comparing Research Attitudes in Down Syndrome and Non-Down Syndrome Decision-Makers”, August 18, 2023.

UCSB Data Science Symposium: **Leader, S.**, Zabback, A., White, B., Yang, T. “Graphing Related Illicit Message Entities Using Network Analysis”, June 5, 2023.

Cal Poly College of Science and Mathematics Research Conference: **Leader, S.**, Chan, K. “Investigating Simulation-Based Curriculum in Introductory Statistics Courses”, May 13, 2022

Cal Poly Frost Summer Undergraduate Research Symposium: **Leader, S.**, Chan, K. “Investigating Simulation-Based Curriculum in Introductory Statistics Courses”, October 1, 2021
